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Thouless' theorem

$$|c'\rangle = \exp \left\{ \sum_{a_1} c_1^a a_1^\dagger a_1 \right\} |c\rangle$$

$$(|c\rangle = |\Phi_c\rangle \quad (\Phi_c^{HF}))$$

$$= |c'\rangle = \prod_{i \leq F} (1 + \sum_{a \geq F} c_i^a a_i^\dagger a_i) |c\rangle$$

$$|c\rangle = a_{i_1}^\dagger a_{i_2}^\dagger \dots a_{i_N}^\dagger |0\rangle$$

$$\begin{aligned}
 |c'\rangle &= \left\{ \left[\left(1 + \sum_a c_{i_1}^a a_a^\dagger a_{i_1} \right) a_{i_1}^\dagger \right] \right. \\
 &\quad \times \left[\left(1 + \sum_a c_{i_2}^a a_a^\dagger a_{i_2} \right) a_{i_2}^\dagger \right] \\
 &\quad \vdots \\
 &\quad \left. \left[\left(1 + \sum_a c_{i_N}^a a_a^\dagger a_{i_N} \right) a_{i_N}^\dagger \right] \right\} \\
 &\quad \times |0\rangle
 \end{aligned}$$

$$= \prod_{i' \leq F} \underbrace{\left(a_{i'}^\dagger + \sum_{a > F} c_i^a a_a^\dagger \right)}_{b_i^\dagger} |0\rangle$$

$$|C'\rangle = \prod_{i \in F} b_i^\dagger |0\rangle$$

$$(|C\rangle = \prod_{i \in F} a_i^\dagger |0\rangle)$$

$$b_i^\dagger = a_i^\dagger + \sum_{a > F} c_a a a_i^\dagger$$

has constraints
(we want $b_i^\dagger$ to have
no restrictions)

Small reminder:

$$|X\rangle = a_X^\dagger |0\rangle$$

$$|P\rangle = a_P^\dagger |0\rangle \quad \wedge \quad a_P^\dagger = \sum_{\lambda} c_{P\lambda} a_\lambda^\dagger$$

Define

$$\tilde{b}_i^+ = \sum_p g_{ip} a_p^+$$

no constraints

$$|\tilde{c}\rangle = \prod_{i \leq F} \tilde{b}_i^+ |0\rangle$$

assume $\underline{\langle c | \tilde{c} \rangle} = \underline{1}$

$$\langle c | \tilde{c} \rangle =$$

$$\begin{aligned}
&= \langle 0 | a_{i_N} a_{i_{N-1}} \dots a_{i_2} a_{i_1} \left(\sum_{p=1}^{i_N} c_{ip} a_p^\dagger \right) \\
&\quad \times \left(\sum_{q=1}^{i_N} g_{i_2 q} a_q^\dagger \right) \\
&\quad \dots \left(\sum_{t=1}^{i_N} g_{i_N t} a_t^\dagger \right) | 0 \rangle
\end{aligned}$$

Two states $i_1 = 1$ and $i_2 = 2$

$$|e\rangle = a_1^\dagger a_2^\dagger |0\rangle$$

$$\begin{aligned}
\langle C | \tilde{C} \rangle &= \langle 0 | a_2 a_1 (g_1 a_1^\dagger + g_2 a_2^\dagger) \\
&\quad \times (g_{21} a_1^\dagger + g_{22} a_2^\dagger) | 0 \rangle = \underline{1}
\end{aligned}$$

$$= \langle 0 | \left\{ g_2 a_1 g_{11} g_{22} a_1^\dagger a_2^\dagger + g_{12} g_2 g_1 a_2^\dagger a_1^\dagger + g_2 a_1 g_{11} \cancel{g_{21}} a_1^\dagger a_1^\dagger + g_2 a_1 g_{22} \cancel{a_2^\dagger} g_{12} a_2^\dagger \right\} | 0 \rangle$$

$$= \langle 0 | \underbrace{g_{12} g_{22} - g_{12} g_{21}}_{\det(g)} | 0 \rangle = 1$$

$$\langle C | \tilde{C} \rangle = 1 = \det(G)$$

$$\sum_k g_{ik} g_{kj}^{-1} = \delta_{ij}$$

$$\sum_j g_{ij}^{-1} g_{jk} = \delta_{ik}$$

$$\sum_i g_{ki}^{-1} \underline{b_i^+} = \sum_i g_{ki}^{-1} \sum_{p=1}^{\infty} g_{ip} a_p^+$$

$$= a_k^+ + \sum_i \sum_{p=n+1}^{\infty} g_{ik}^{-1} s_{ip} a_p^+$$

$$C_{kp} = \sum_{i \leq F} g_{ki}^{-1} s_{ip}$$

$$a_k^+ + \sum_{p=n+1}^{\infty} C_{kp} a_p^+ \quad k \leq F$$

nothing but

$$b_n^+ = a_n^+ + \sum_{a > F} C_n^a a_a^+$$

$$|c\rangle = \prod_{i \in F} b_i^\dagger |0\rangle =$$

$$\prod_{i \in F} b_i^\dagger |0\rangle$$

$$b_i^\dagger = a_i^\dagger + \sum_{q \in F} c_i^a a_q^\dagger$$

$$|c'\rangle = \exp\left\{\sum_{a \in F} c_i^a a_a^\dagger \xi_i\right\} |c\rangle$$

Stability of HF eqs.

$$\text{is } \langle \Phi_0^{\text{HF}} | H | \Phi_0^{\text{HF}} \rangle \text{ a}$$

better variational
minimum?

$$|\Phi_0^{\text{HF}}\rangle = C |\Phi_0\rangle$$

$$\langle \Phi_0^{\text{HF}} | H | \Phi_0^{\text{HF}} \rangle \leq \langle \Phi_0 | H | \Phi_0 \rangle$$

$$|\Phi_0^{\text{HF}}\rangle = |C\rangle$$

$$\langle c | \hat{H} | c \rangle = E_0^{HF} = E_0^{Ref}$$

$$|c'\rangle = |c\rangle + |\delta c\rangle$$

$$= \exp \left\{ \sum_{q_i} \delta c_i a_{q_i}^\dagger a_{q_i} \right\} |c\rangle$$

$$\frac{\langle c' | \hat{H} | c' \rangle}{\langle c' | c' \rangle} \geq \langle c | \hat{H} | c \rangle$$

which are the conditions
which are needed to
satisfy this equation?

Denominator

$$\langle c' | c' \rangle = 1 + \sum_{\substack{a, b \\ i, j}} \delta C_i^* \delta C_j$$

$$\times \underbrace{\langle c | a_i^\dagger a_a a_i^\dagger a_j | c \rangle}_{\delta a_i \delta a_j} + \dots$$

$$= 1 + \sum_{a, i} |\delta C_i^a|^2 + O(\delta C^3)$$

$$\approx 1 + \sum_{a, i} |\delta C_i^a|^2$$

$$\langle c' | \hat{H} | c' \rangle = \underbrace{\langle c | \hat{H} | c \rangle}_{E_0^{HF}}$$

$$|c'\rangle = |c\rangle + |\delta c\rangle$$

$$+ \underbrace{\langle \delta c | \hat{H} | c \rangle}_{\text{IP12}} + \underbrace{\langle c | \hat{H} | \delta c \rangle}_{\text{IP12}} + \langle \delta c | \hat{H} | \delta c \rangle$$

$$\langle \delta c | \hat{H} | \delta c \rangle = ?$$

$$|c'\rangle = \exp \left\{ \sum_{\mathbf{q}_i} \delta c_{\mathbf{q}_i} a_{\mathbf{q}_i}^\dagger a_{\mathbf{q}_i} \right\} |c\rangle$$

To 2nd order in δC

$$(i) \sum_{ab} \delta C_n^{*a} \delta C_j^b$$

$$+ \langle c | a_n^\dagger a_a \hat{H} a_b^\dagger a_j | c \rangle$$

$$(i') \frac{1}{2} \sum_{ab} \delta C_n^a \delta C_j^b \langle c | \hat{H} a_a^\dagger a_i a_b^\dagger a_j | c \rangle$$

$$(i'') \frac{1}{2} \sum_{ab} \delta C_n^{*a} \delta C_j^{*b} \langle c | a_j^\dagger a_b a_n^\dagger a_a \hat{H} | c \rangle$$

$$\hat{H} = E_0^{HF} + \hat{F}_N + \hat{V}_N$$

First term (i)

$$a) \sum_{a_i} E_0^{HF} \langle c | \overbrace{a_i^\dagger a_a a_b^\dagger a_j}^{+ +} | c \rangle \times \delta c_i^{*a} \delta c_j^b$$

$$= E_0^{HF} \sum_{a_i} |\delta c_i^a|^2$$

$$b) \sum_{a_i} \hat{F}_N \sum_{pq} \delta c_i^{*a} \delta c_j^b \langle c | \overbrace{a_i^\dagger a_a a_p^\dagger a_q}^{+ +} \underbrace{a_r^\dagger a_{j'}}_{\delta_{ij}} | c \rangle \times \langle p | f | q \rangle$$

$\delta_{i'q}$ $\delta_{a,b}$ $\delta_{p,j'}$

δ_{ap} $\delta_{q,b}$

$$\langle a | \hat{f} | b \rangle \delta_{ab} = \epsilon_a^{HF}$$

$$\sum_{a \neq i} |\delta c_a^a|^2 (\epsilon_a^{HF} - \epsilon_i^{HF})$$

c)

$$\sum_{\substack{a \neq i \\ r_j}} \frac{1}{4} \sum_{p \neq r_j} \langle p \hat{g} | v | r_j \rangle_{A_j} \delta c_n^{*a} \delta c_j^b$$

$$\langle C | \underbrace{a_n^\dagger a_a a_p^\dagger a_q^\dagger}_{\text{}} a_r a_n a_b^\dagger a_j | C \rangle$$

$$- \frac{1}{4} \sum_{\substack{a \neq i \\ r_j}} \delta c_n^{*a} \delta c_j^b \langle a_j | v | r_i \rangle_{A_j}$$

+ 3 more sets

(i')

$$\sum_{a_i} |\delta c_{i'}^a|^2 (E_0^{HF} + (\epsilon_a^{HF} - \epsilon_{i'}^{HF}))$$

$$+ \sum_{\substack{a \\ r}} \delta c_{i'}^a \delta c_j^r \langle a_j' | \vec{r} | b_{i'} \rangle_{AS}$$

(1p1h block in FCI
with a HF basis
 $\langle 1p1h | \hat{H} | 1p1h \rangle$)

(ii)

$$\sum_{\substack{ab \\ ij}} \delta c_a^i \delta c_j^b \langle ij | v | ab \rangle_{AJ}$$

(iii)

$$\sum_{\substack{ab \\ ij}} \delta c_a^{*i} \delta c_j^{*b} \langle ab | v | ij \rangle_{AJ}$$

$$\langle c | H | 2p24 \rangle$$

$$\text{and } \langle 2p24 | H | c \rangle$$

$$\Rightarrow \langle c' | H | c' \rangle =$$

$$= E_0^{HF} \left(1 + \sum_{a_i} |\delta c_i^a|^2 \right)$$

$$+ \Delta E + O(\delta c^3)$$

$$\frac{\langle c' | H | c' \rangle}{\langle c' | c' \rangle} \approx E_0^{HF} + \frac{\Delta E}{1 + \sum_{a_i} |\delta c_i^a|^2}$$

$$I = \{a_i\} \quad \text{1p1h state}$$

$$J = \{b_j\} \quad \text{--- 1 ---}$$

(i')

$$\sum_{\substack{ab \\ ij}} \delta c_i^{*a} \delta c_j^b \left\{ \left[\epsilon_a^{HF} \delta_{ab} \delta_{ij}' - \epsilon_i^{HF} \times \delta_{ab} \delta_{ij}' \right] + \langle a_j' | v | i a \rangle_{AS} \right\}$$

$$\langle a_j' | v | i a \rangle_{AS} = A_{ai, bj'}$$

$$\Delta \epsilon_{IJ} = (\epsilon_a^{HF} - \epsilon_i^{HF}) \delta_{ab} \delta_{ij}' = A_{IJ}$$

$$\delta c_i^{*a} = \delta c_I^{*} \quad \wedge \quad \delta c_j^b = \delta c_J$$

(ii) contains $\langle ij | H | al \rangle$

$$= D_{IJ}$$

$$(iii) \langle al | H | ij \rangle = B_{IJ}^*$$

$$x^T A x = \sum_{i,j} x_i a_{ij} x_j$$

$$\Delta E = \sum_{IJ} \delta C_I^* \delta C_J (\Delta \varepsilon_{IJ} \delta_{IJ} + A_{IJ}) + \sum_{IJ} \delta C_I \delta C_J B_{IJ}^* +$$

$$+ \sum_{I,J} \delta c_I^* \delta c_J B_{IJ}$$

$$\Delta E = \frac{1}{2} \langle X | M | X \rangle$$

$$X = \begin{bmatrix} \delta c \\ \delta c^* \end{bmatrix}$$

$$M = \begin{bmatrix} \Delta E + A & B \\ B^* & \Delta E + A^* \end{bmatrix}$$

we want to prove that $\langle C | H | C \rangle$ is a variational minimum $\Rightarrow$

$$\Delta E \geq 0 \quad \Rightarrow$$

M has to be positive semi-definite $\Rightarrow$

λ of M are all $\lambda \geq 0$

A necessary (but not sufficient) condition is that the sum over $\sum_{I \neq J} (\Delta E_{IJ} + A_{IJ}) \geq 0$